

Models: Fitting and Evaluation

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Microplastics

Total model

```

Family: nbinom2 ( log )
Formula:
particles ~ category + bottle_material + source_type + adv_filtration
Zero inflation:      ~.
Data: mp.total

```

AIC	BIC	logLik	-2*log(L)	df.resid
414.6	435.9	-196.3	392.6	40

Dispersion parameter for nbinom2 family (): 1.22

Conditional model:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	3.1512	0.5717	5.512	3.55e-08	***
categoryBottled	1.9115	0.6683	2.860	0.00423	**
bottle_materialPET	-1.5790	0.3624	-4.358	1.32e-05	***
bottle_materialHDPE	NA	NA	NA	NA	
source_typeSpring	-0.2986	0.4329	-0.690	0.49037	
adv_filtrationYes	0.2620	0.4951	0.529	0.59667	

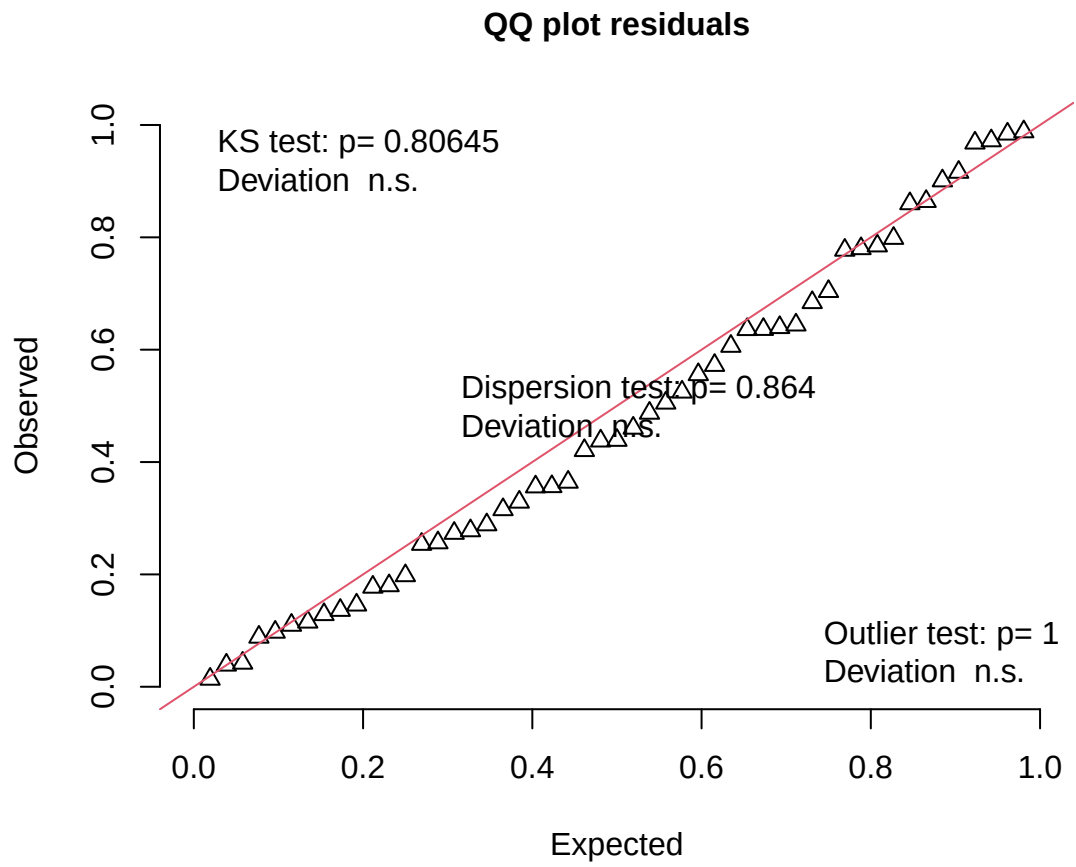
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

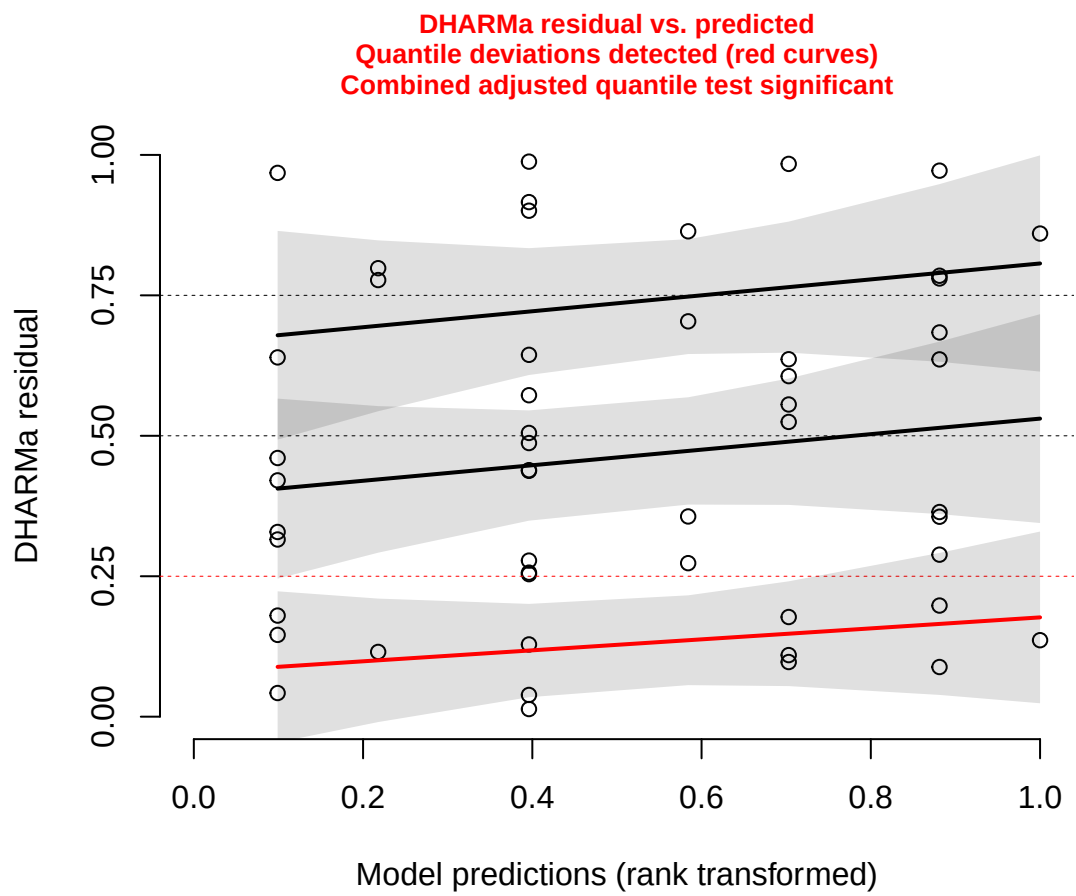
Zero-inflation model:

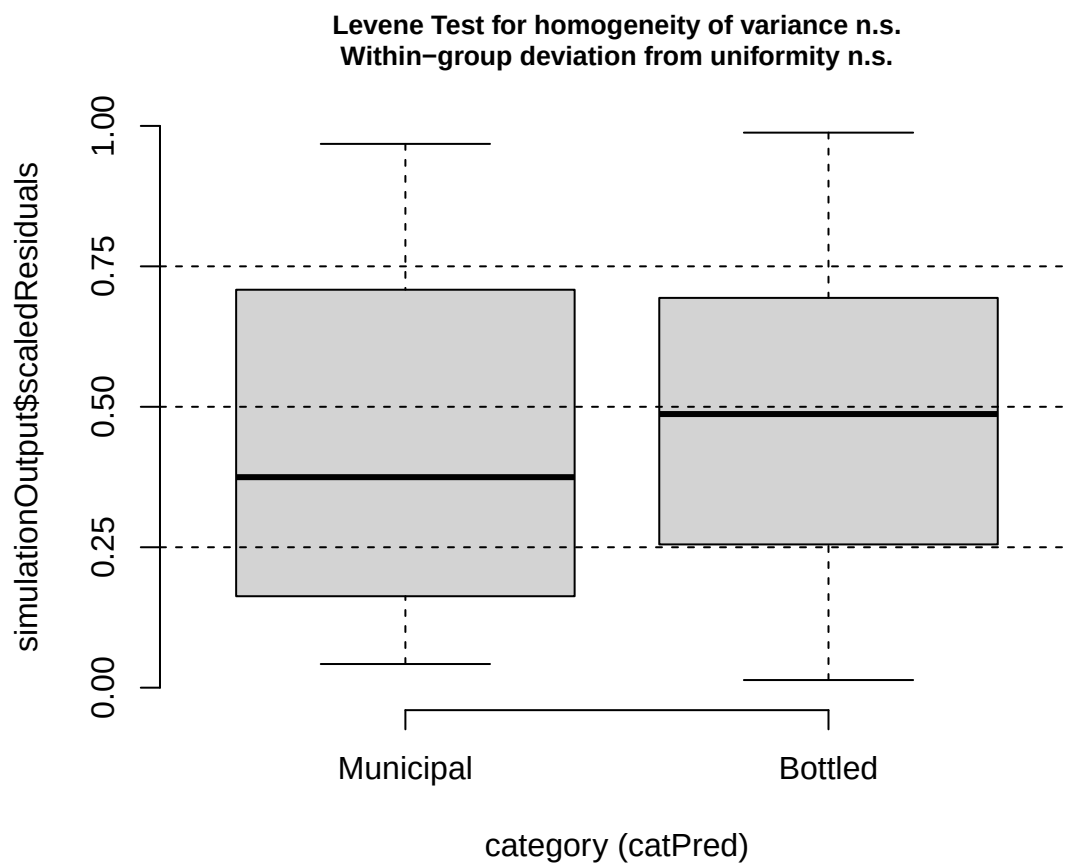
	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	1.2563	0.7636	1.645	0.0999	.
categoryBottled	-3.6909	1.5543	-2.375	0.0176	*
bottle_materialPET	1.0085	1.2452	0.810	0.4180	
bottle_materialHDPE	NA	NA	NA	NA	
source_typeSpring	-0.0125	1.1606	-0.011	0.9914	
adv_filtrationYes	-0.6730	1.1230	-0.599	0.5490	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Model evaluation







Bottled model

```
Family: nbinom2 ( log )
Formula:          particles ~ bottle_material + source_type + adv_filtration
Zero inflation:    ~.
Data: mp.bottled.total
```

AIC	BIC	logLik	-2*log(L)	df.resid
369.5	384.5	-175.8	351.5	30

Dispersion parameter for nbinom2 family (): 1.19

Conditional model:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	3.3120	0.4030	8.218	< 2e-16 ***
bottle_materialHDPE	1.6313	0.3646	4.474	7.67e-06 ***
source_typeSpring	-0.1514	0.4318	-0.351	0.726
adv_filtrationYes	0.5816	0.5534	1.051	0.293

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Zero-inflation model:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-19.3246	7037.5151	-0.003	0.998
bottle_materialHDPE	-0.7074	1.3152	-0.538	0.591
source_typeSpring	17.7934	7037.5151	0.002	0.998
adv_filtrationYes	18.1784	7037.5152	0.003	0.998

The large standard errors in the zero-inflated component are because source_type and adv_filtration are very collinear.

	No	Yes
Non-spring	6	8
Spring	25	0

	FALSE	TRUE
Non-spring	12	2
Spring	20	5

	FALSE	TRUE
No	25	6
Yes	7	1

```
# A tibble: 1 x 8
# Groups:   id, category, bottle_material, source_type [1]
  id      category bottle_material source_type adv_filtration particles    nd
<chr>    <fct>    <fct>          <fct>      <fct>          <dbl> <dbl>
1 Brand C-1 Bottled PET             Non-spring Yes             157     0
# i 1 more variable: detect <dbl>
```

So, we do not look at the zero-inflated portion of the model, as the estimates and inference are not reliable.

We check that neither variable is significant on its own:

```
Family: nbinom2 ( log )
Formula:          particles ~ bottle_material + source_type
Zero inflation:    ~.
Data: mp.bottled.total
```

AIC	BIC	logLik	-2*log(L)	df.resid
368.6	380.2	-177.3	354.6	32

Dispersion parameter for nbinom2 family (): 1.13

Conditional model:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	3.6468	0.2814	12.959	< 2e-16 ***
bottle_materialHDPE	1.5392	0.3702	4.158	3.21e-05 ***
source_typeSpring	-0.4477	0.3641	-1.229	0.219

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Zero-inflation model:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.8168	0.8669	-2.096	0.0361 *
bottle_materialHDPE	-0.9295	1.2716	-0.731	0.4648
source_typeSpring	0.3336	1.0769	0.310	0.7567

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
Family: nbinom2 ( log )
Formula:          particles ~ bottle_material + adv_filtration
Zero inflation:    ~.
Data: mp.bottled.total
```

AIC	BIC	logLik	-2*log(L)	df.resid
367.4	379.0	-176.7	353.4	32

Dispersion parameter for nbinom2 family (): 1.16

Conditional model:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	3.1908	0.2386	13.371	< 2e-16	***
bottle_materialHDPE	1.6317	0.3685	4.428	9.5e-06	***
adv_filtrationYes	0.7018	0.4513	1.555	0.12	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

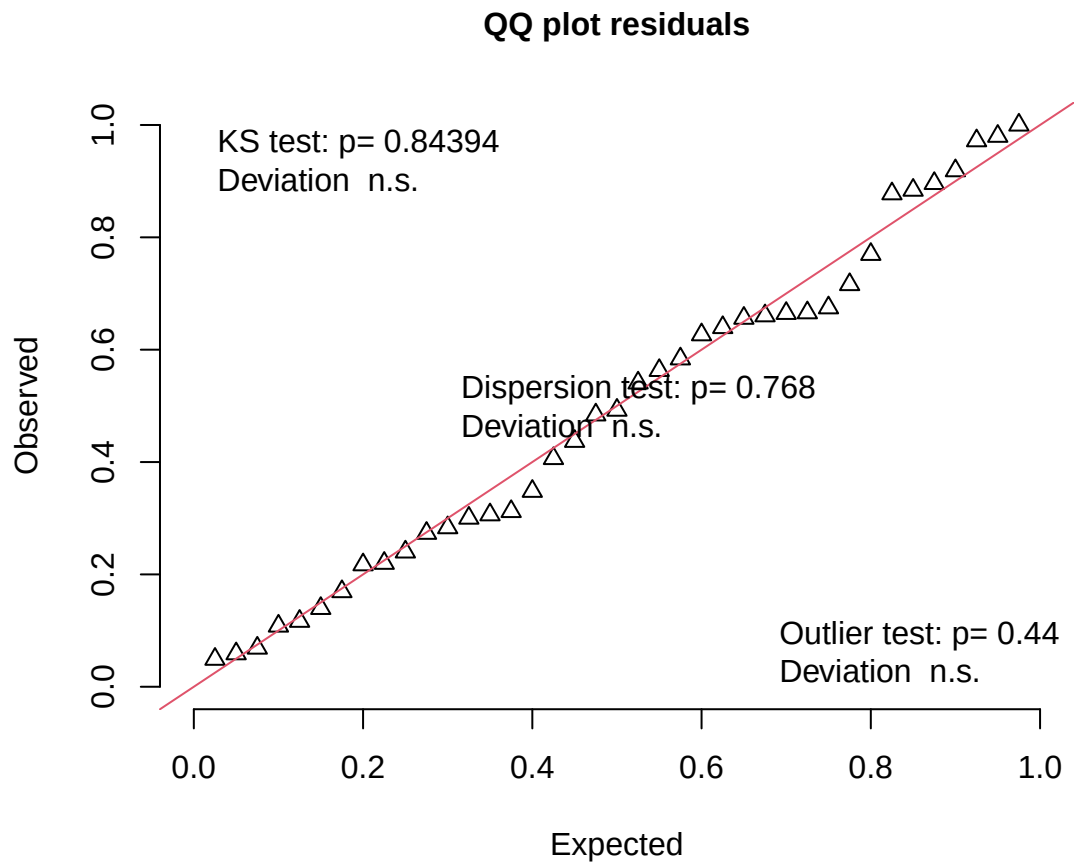
Zero-inflation model:

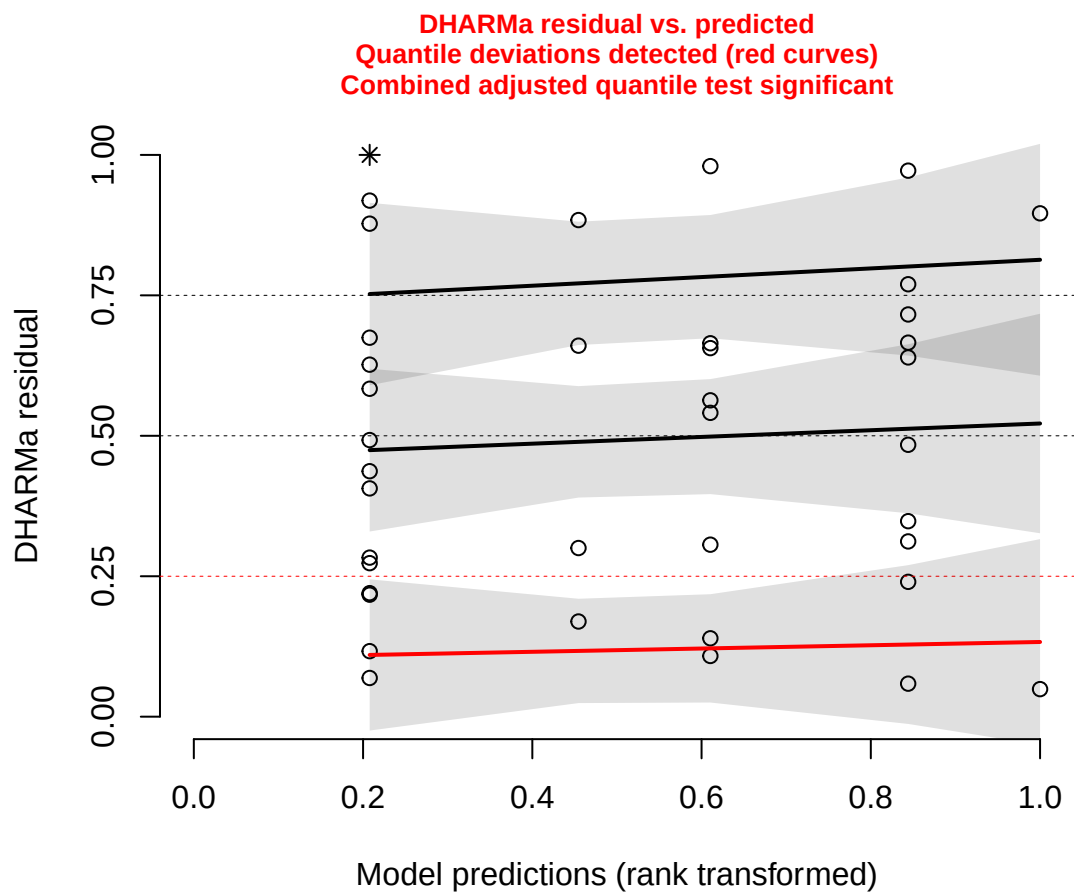
	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-1.8664	0.7579	-2.463	0.0138	*
bottle_materialHDPE	-0.5849	1.3280	-0.440	0.6596	
adv_filtrationYes	0.7163	1.1332	0.632	0.5273	

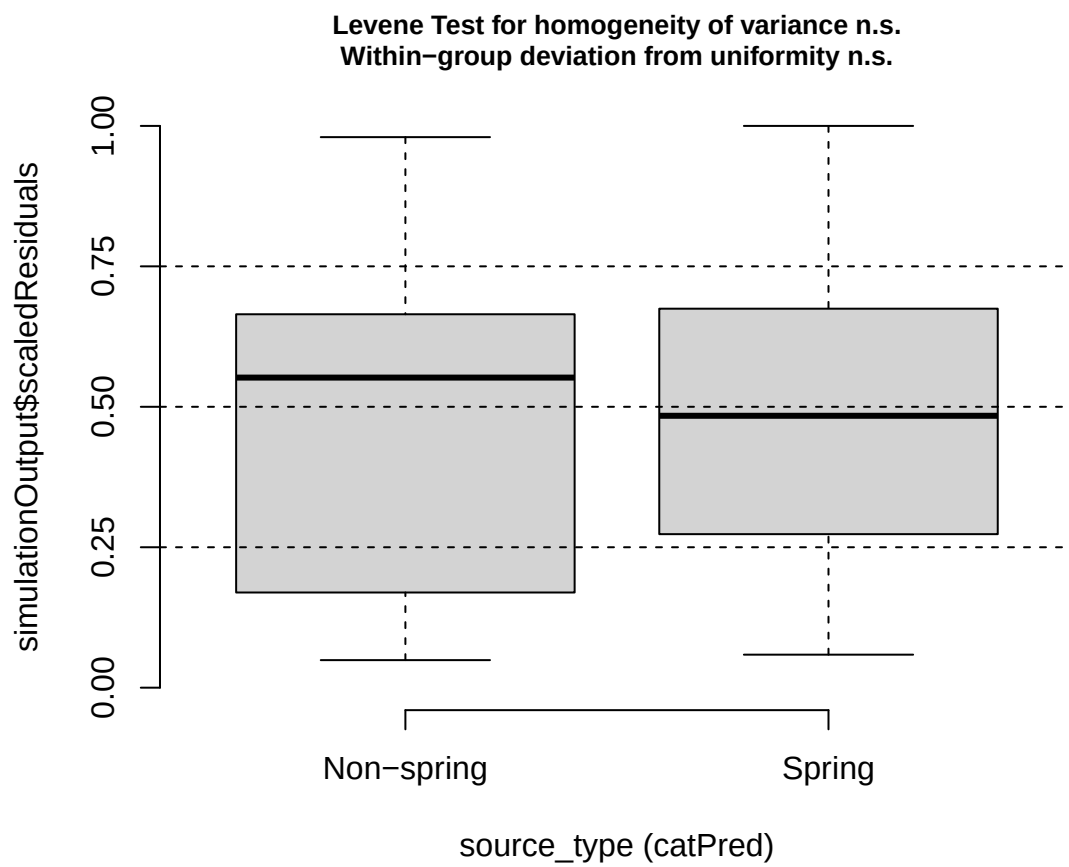
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

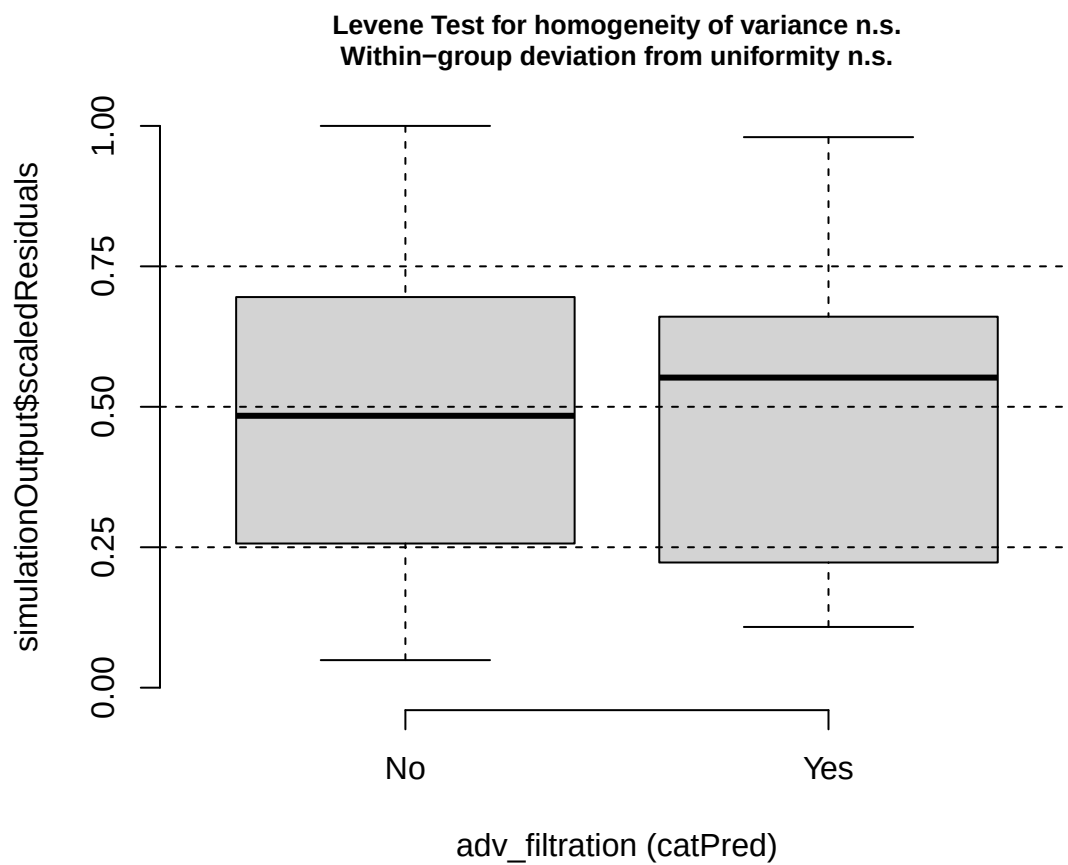
Because neither variable is significant in the zero-inflated component on its own, we are reasonably confident there is no difference in detection probability by either source type or advanced filtration. To report p-values, we instead report the Fisher exact test p-values, which don't control for other variables, as the p-values from the zero-inflated portion of the model are not reliable. See below for the Fisher exact tests.

Model evaluation









Fisher exact tests

Probability of detection: source

[1] 0.84

[1] 0.8571429

	0	1
Non-spring	2	12
Spring	4	21

Exact Two-Sample Fisher-Pitman Permutation Test

```
data: detect by factor(source_type) (Non-spring, Spring)
Z = 0.1405, p-value = 1
alternative hypothesis: true mu is not equal to 0
```

Probability of detection: bottle material

	0	1
PET	5	22
HDPE	1	11

Exact Two-Sample Fisher-Pitman Permutation Test

```
data: detect by factor(bottle_material) (PET, HDPE)
Z = -0.80316, p-value = 0.645
alternative hypothesis: true mu is not equal to 0
```

Probability of detection: advanced filtration

```
[1] 0.75
```

```
[1] 0.8709677
```

	0	1
No	4	27
Yes	2	6

Exact Two-Sample Fisher-Pitman Permutation Test

```
data: detect by factor(adv_filtration) (No, Yes)
Z = 0.83455, p-value = 0.5834
alternative hypothesis: true mu is not equal to 0
```

PE model

```
Family: nbinom2 ( log )
Formula:          particles ~ bottle_material + source_type + adv_filtration
Zero inflation:    ~.
Data: mp.bottled.pe
```

AIC	BIC	logLik	-2*log(L)	df.resid
325.6	340.6	-153.8	307.6	30

Dispersion parameter for nbinom2 family (): 0.897

Conditional model:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	4.0518	0.4965	8.161	3.34e-16 ***
bottle_materialPET	-1.4517	0.4692	-3.094	0.00198 **
source_typeSpring	0.5429	0.5590	0.971	0.33144
adv_filtrationYes	1.1332	0.7293	1.554	0.12026

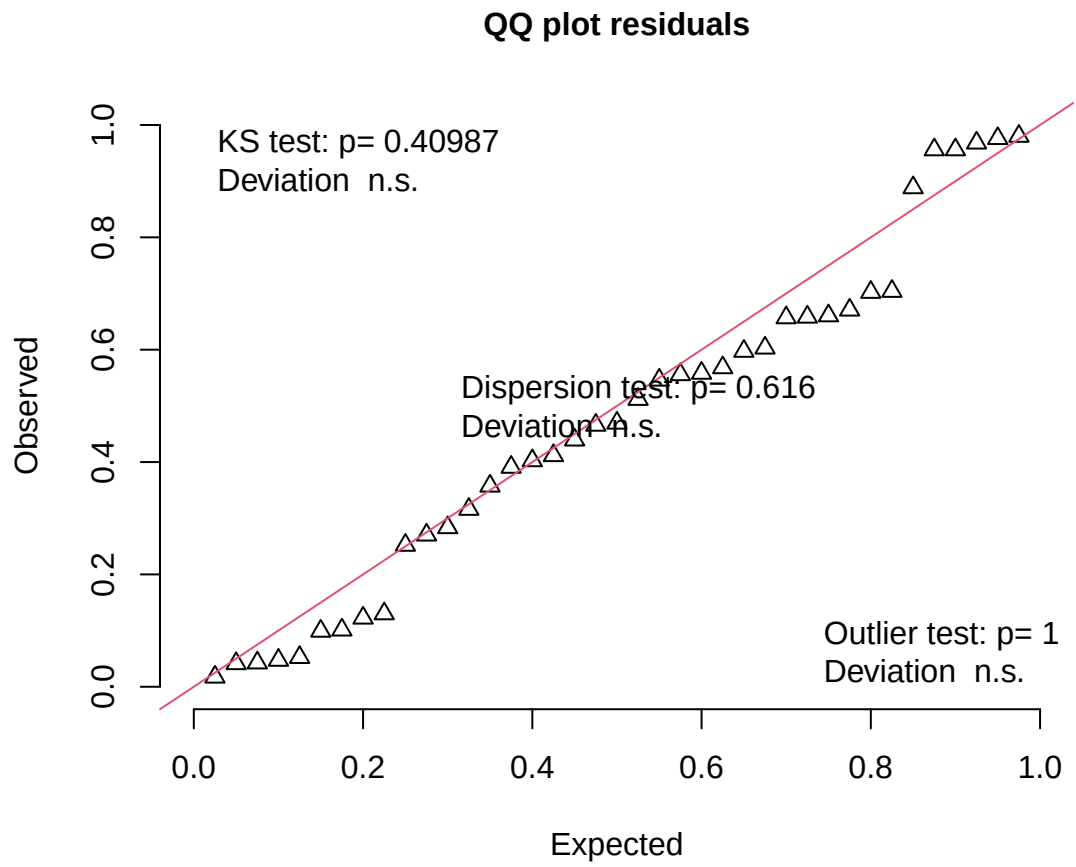
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

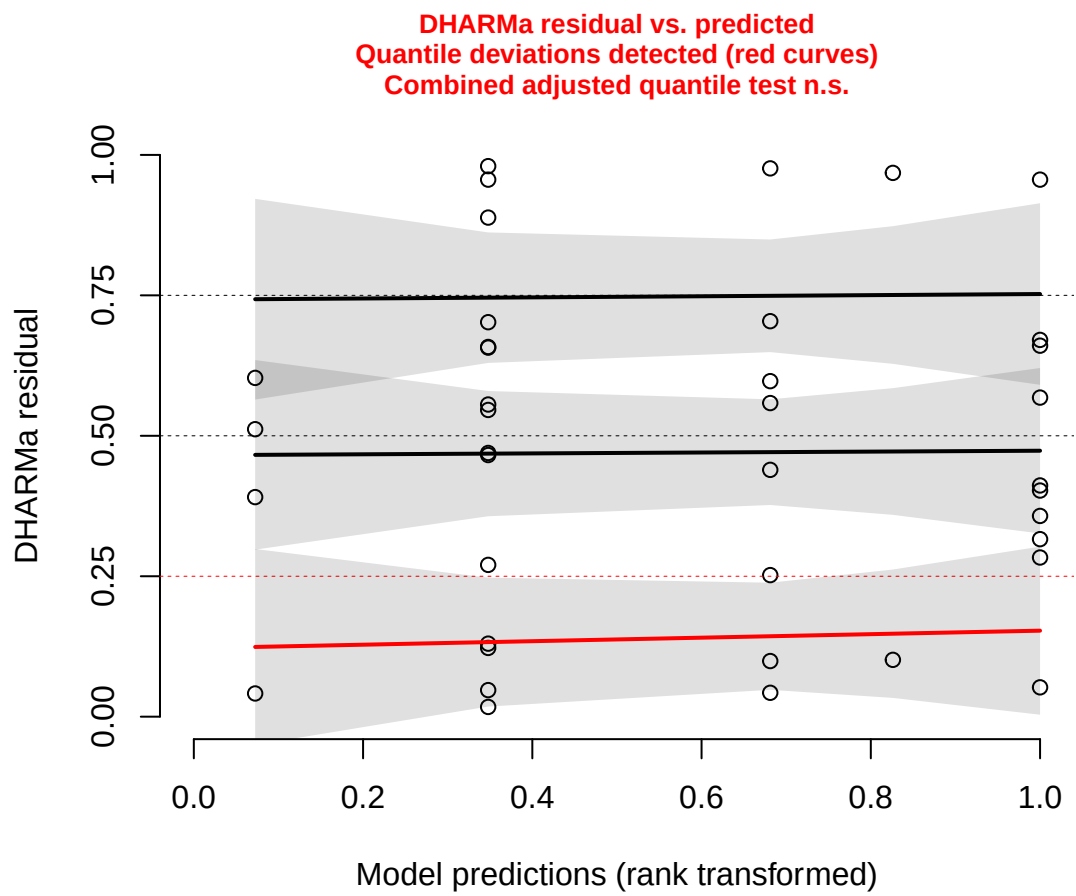
Zero-inflation model:

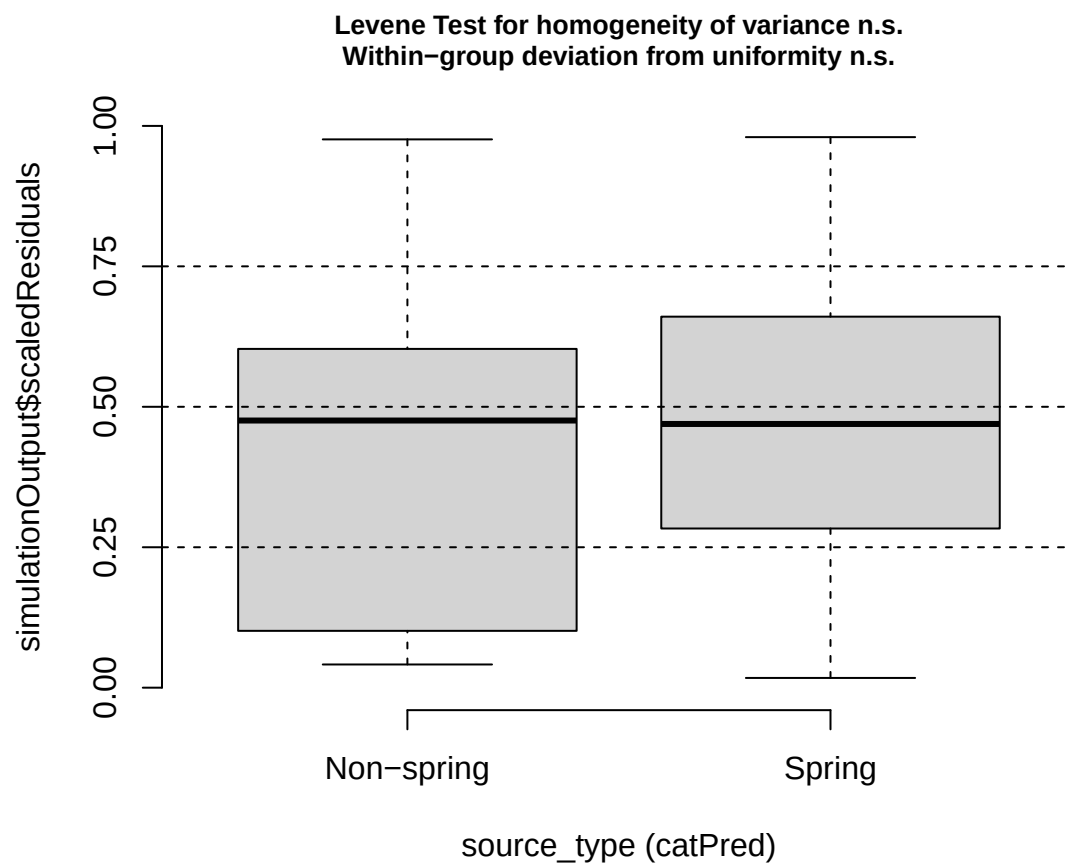
	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-3.8033	2.0454	-1.859	0.063 .
bottle_materialPET	2.1828	1.3356	1.634	0.102
source_typeSpring	1.3814	1.7203	0.803	0.422
adv_filtrationYes	0.3883	1.8810	0.206	0.836

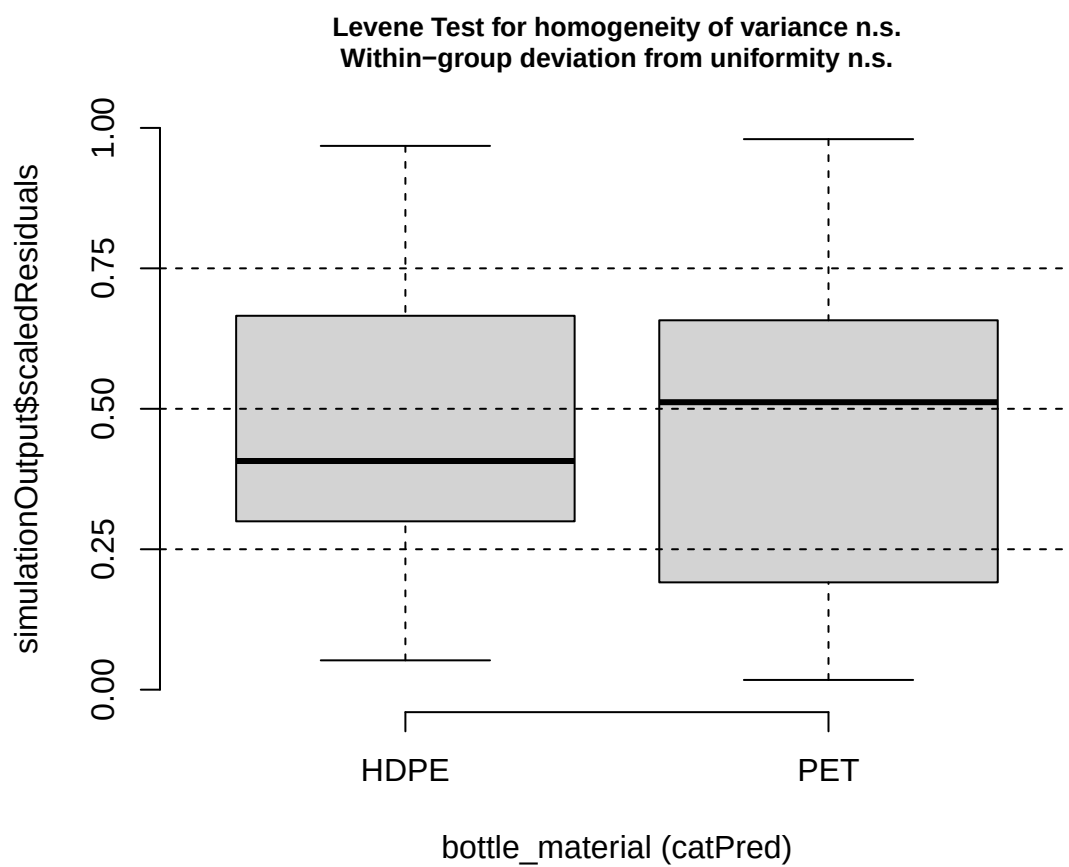
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

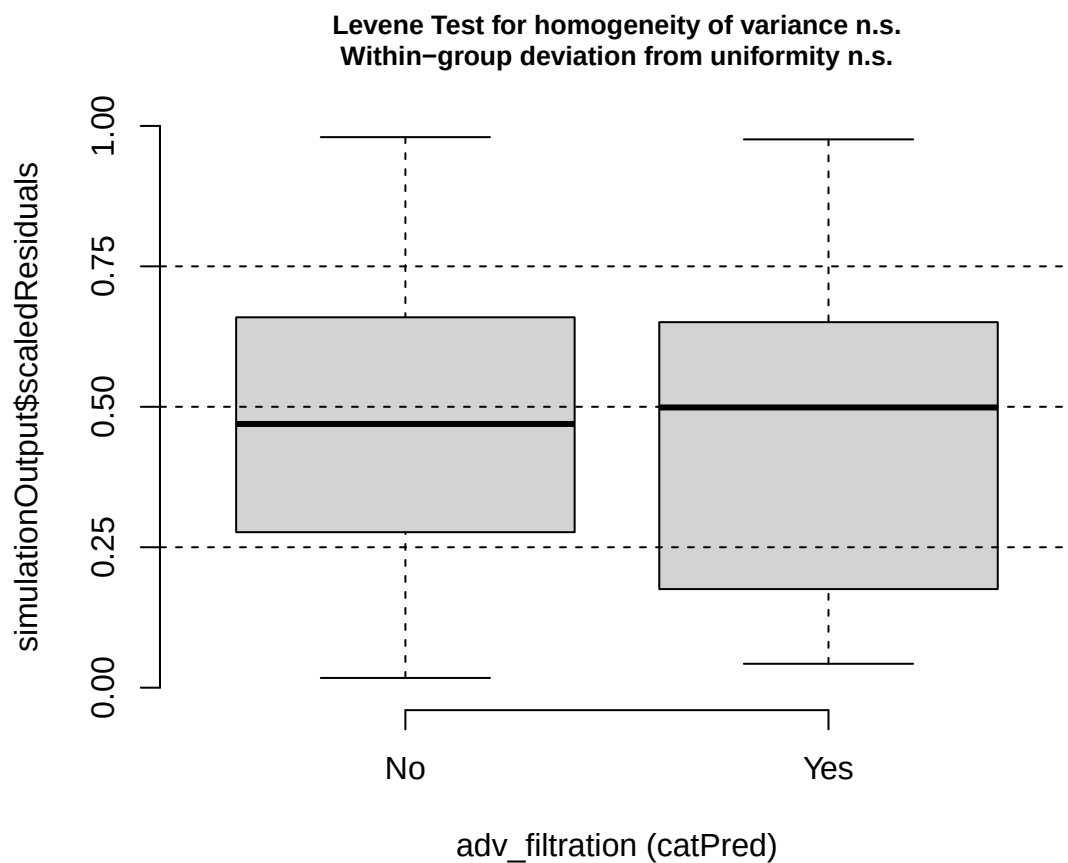
Model evaluation











PET model

```
Family: nbinom2 ( log )
Formula:          particles ~ source_type + bottle_material + adv_filtration
Zero inflation:    ~.
Data: mp.bottled.pet
```

AIC	BIC	logLik	-2*log(L)	df.resid
174.4	189.3	-78.2	156.4	30

Dispersion parameter for nbinom2 family (): 3.49

Conditional model:

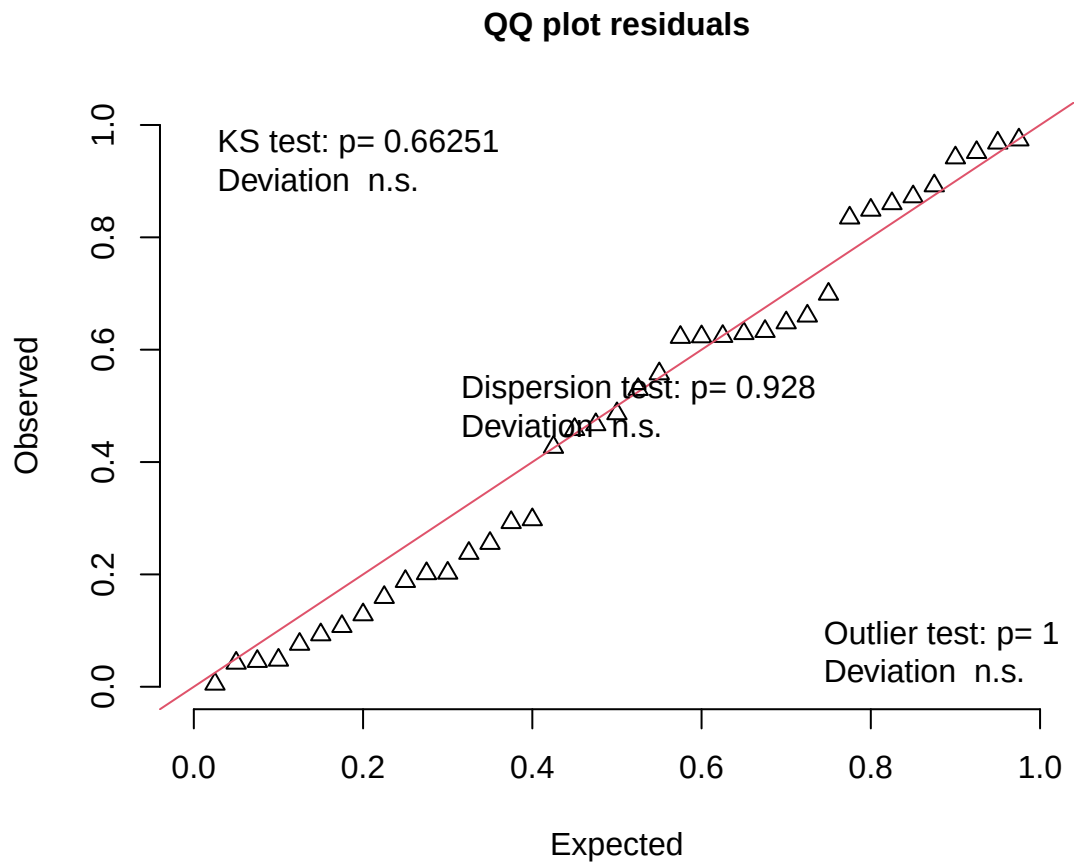
	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	3.2052	0.4468	7.174	7.3e-13 ***
source_typeSpring	0.1953	0.4714	0.414	0.67871
bottle_materialPET	-1.0995	0.3512	-3.131	0.00174 **
adv_filtrationYes	0.4547	0.5932	0.767	0.44333

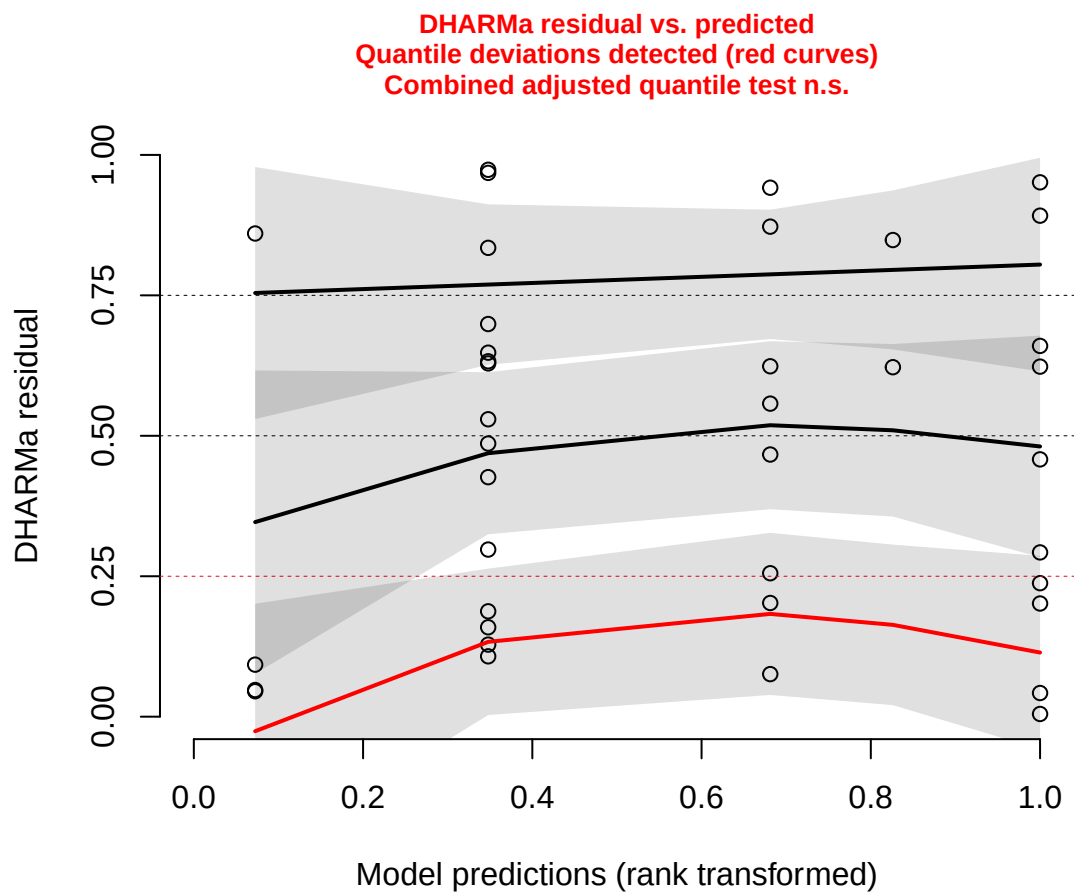
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

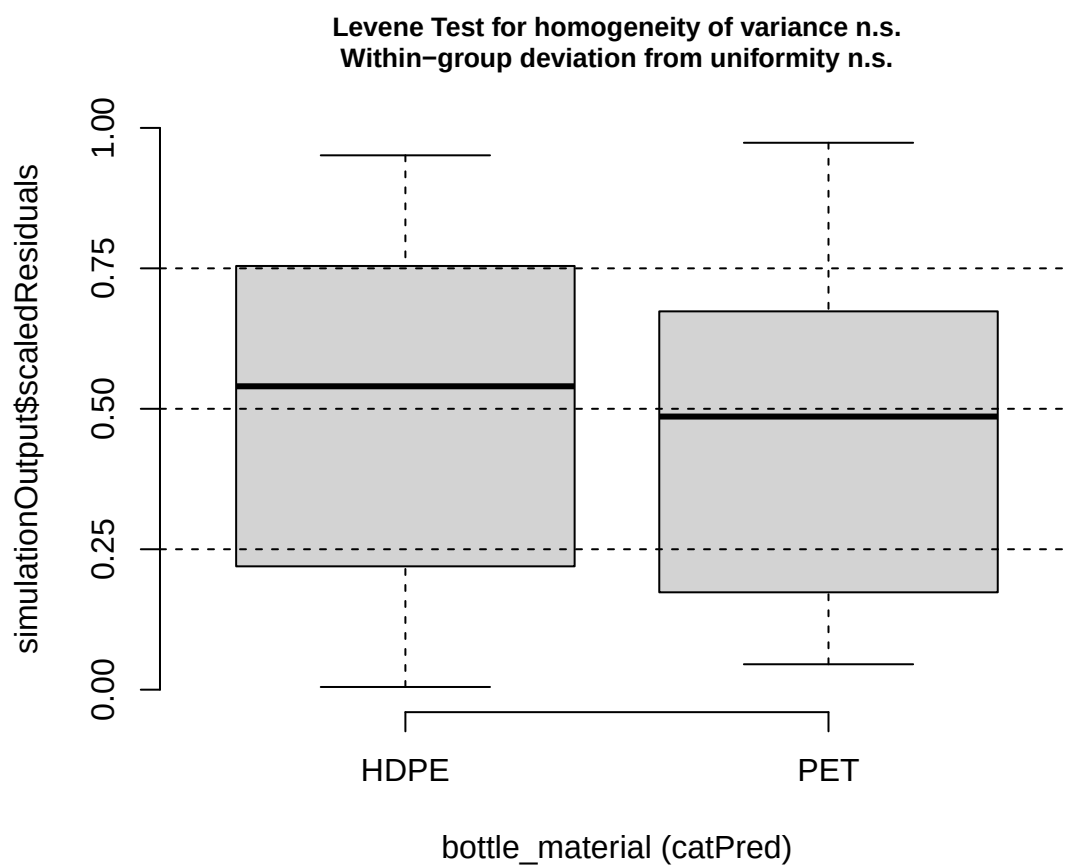
Zero-inflation model:

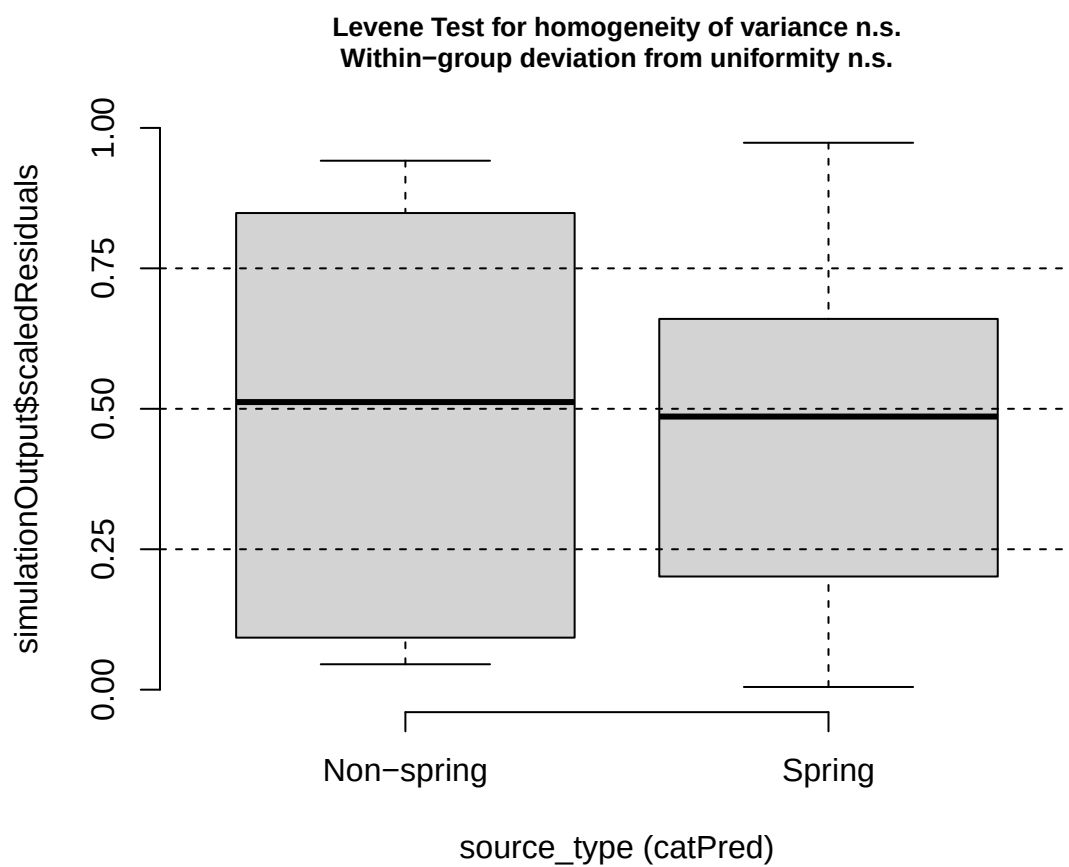
	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.5609	1.0008	0.560	0.575
source_typeSpring	-0.2691	0.9643	-0.279	0.780
bottle_materialPET	0.1761	0.7583	0.232	0.816
adv_filtrationYes	-0.2334	1.1691	-0.200	0.842

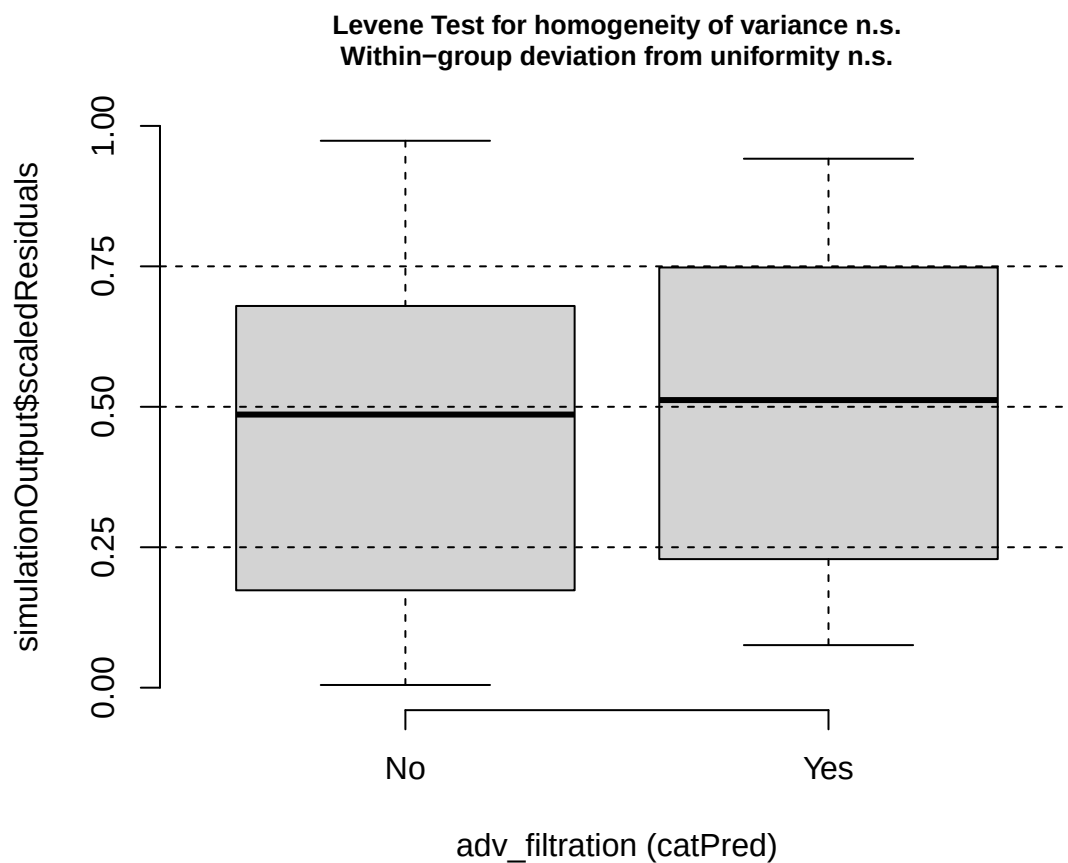
Model evaluation











PFAS

Category and filtration

```
Family: Gamma ( log )
Formula:      concentration ~ category + adv_filtration
Zero inflation: ~.
Data: pfas.total
```

AIC	BIC	logLik	-2*log(L)	df.resid
514.4	528.1	-250.2	500.4	45

Dispersion estimate for Gamma family (σ^2): 0.35

Conditional model:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	5.5472	0.1233	44.99	< 2e-16 ***
categoryMunicipal	0.8109	0.2427	3.34	0.000835 ***
adv_filtrationYes	0.5391	0.4003	1.35	0.178101

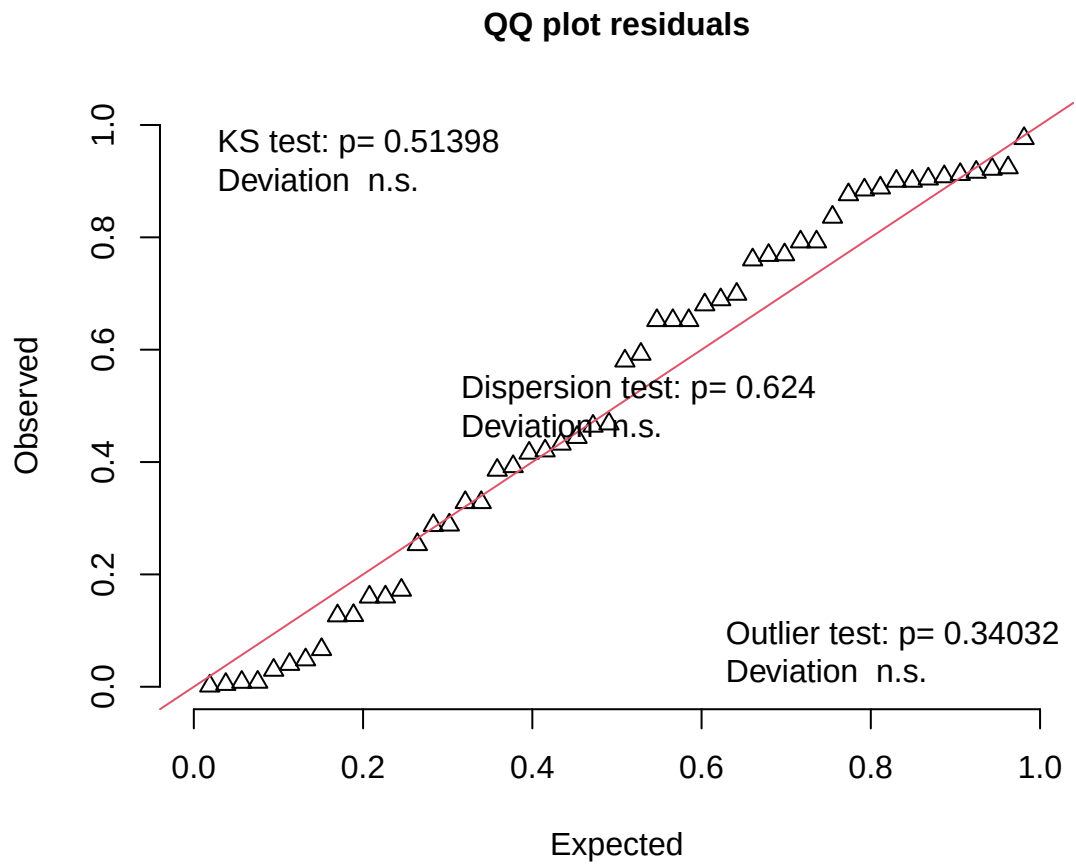
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

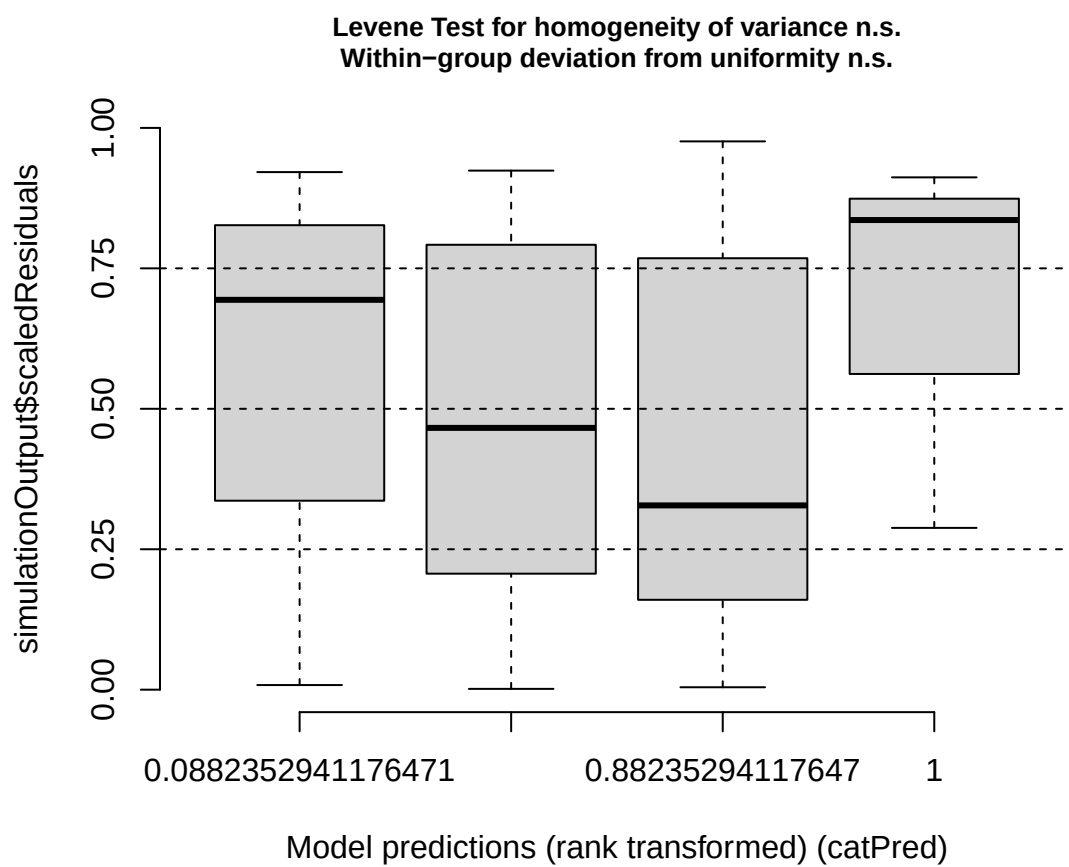
Zero-inflation model:

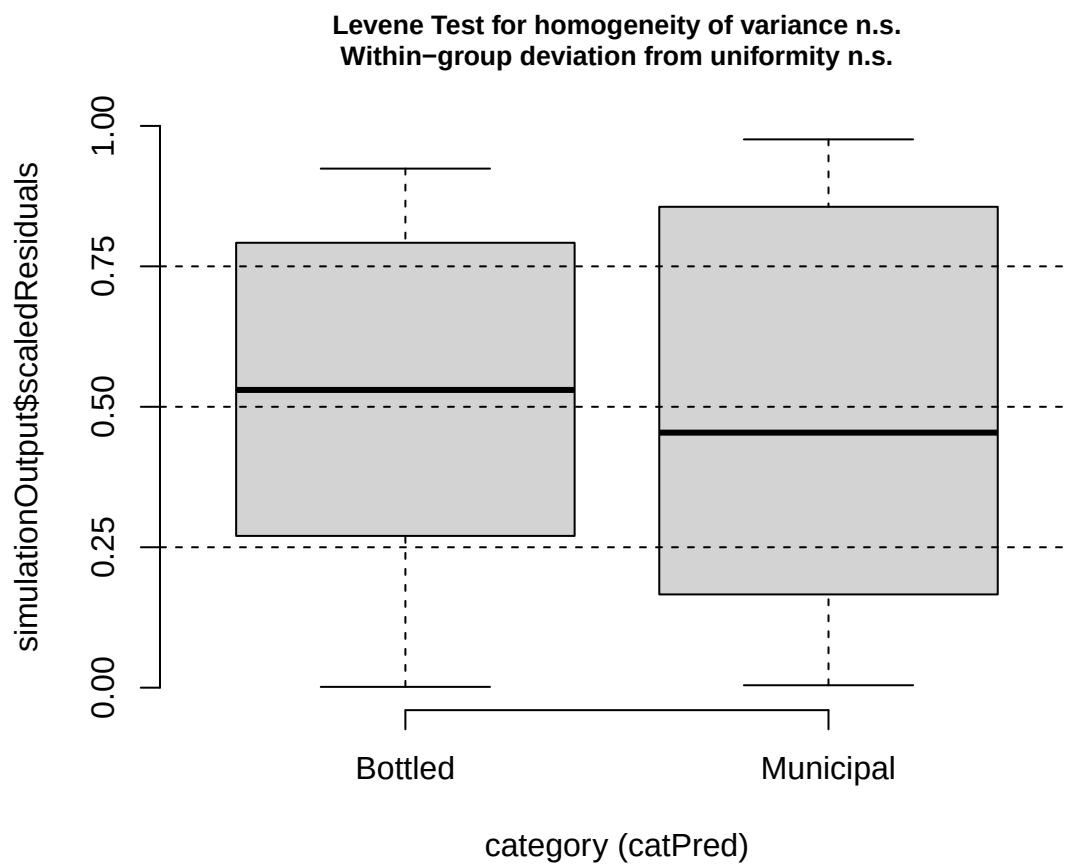
	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.8136	0.3801	-2.140	0.0323 *
categoryMunicipal	-3.1237	1.4194	-2.201	0.0278 *
adv_filtrationYes	2.9727	1.1173	2.660	0.0078 **

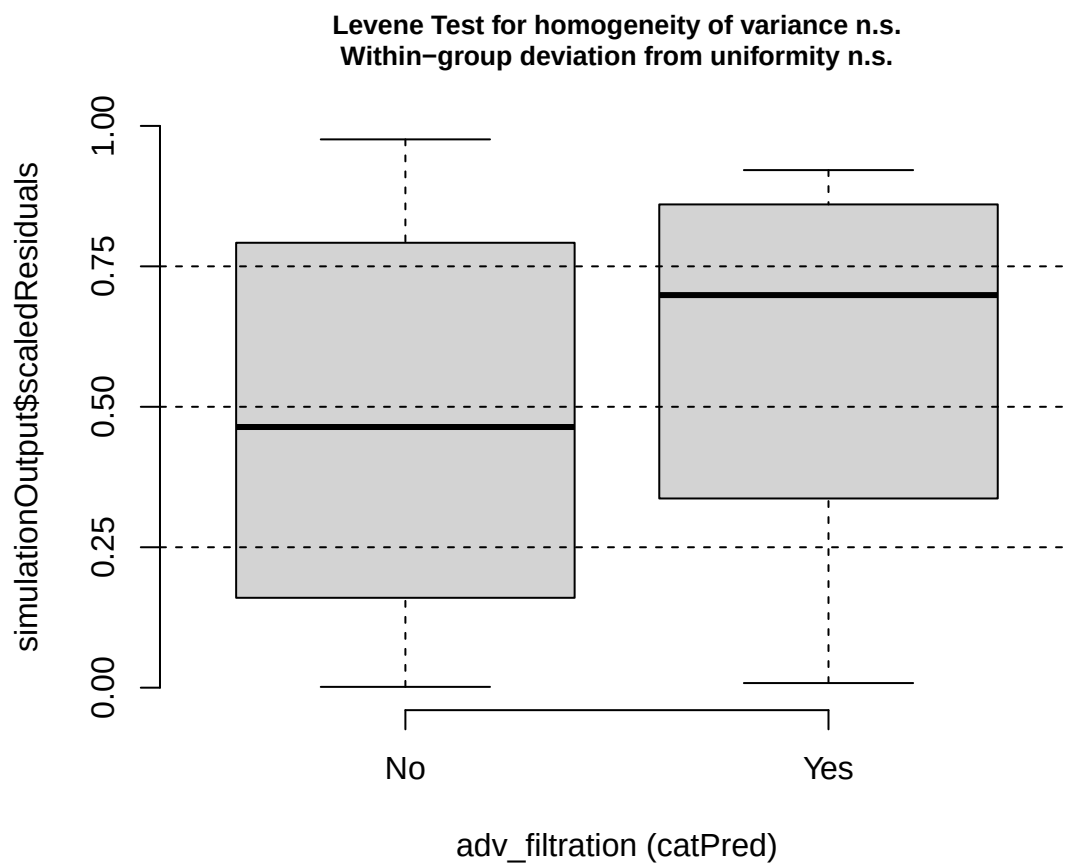
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Model evaluation









Probability of detection: source

```
# A tibble: 2 x 5
  source_type num.detect num.zero total percent
<fct>         <int>    <int> <int>   <dbl>
1 Non-spring      0      13    13      0
2 Spring          23       4    27  85.2
```

```
      0  1
Non-spring 13  0
Spring      4 23
```

Exact Two-Sample Fisher-Pitman Permutation Test

data: detect by factor(source_type) (Non-spring, Spring)
Z = -5.0404, p-value = 1.978e-07
alternative hypothesis: true mu is not equal to 0

Probability of detection: advanced filtration

	0	1
No	10	31
Yes	8	3

	Non-spring	Spring
No	14	27
Yes	11	0

	Bottled	Municipal
No	32	9
Yes	8	3

Exact Two-Sample Fisher-Pitman Permutation Test

data: detect by factor(adv_filtration) (No, Yes)
Z = 2.9633, p-value = 0.004812
alternative hypothesis: true mu is not equal to 0